# Densely Connected AutoEncoders for Image Compression

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## ABSTRACT

Image compression, which is a type of data compression applied to digital images, has been a fundamental research topic for many decades. Recent image techniques produce very large amounts of data, which may make it prohibitive to storage and communications of image data without the use of compression. However, the traditional compression methods, such as JPEG, may introduce the compression artefact problems. Recently, deep learning has achieved great success in many computer vision tasks and is gradually being used in image compression. To solve the compression atrefact problem, in this paper, we present a lossy image compression architecture, which utilizes the advantages of the existing deep learning methods to achieve a high coding efficiency. We design a densely connected autoencoder structure for lossy image compression. Firstly, we design a densely autoencoder structure to get richer feature information from image which can be helpful for compression. Secondly, we design a U-net like network to decrease the distortion caused by compression. Finally, an improved binarizer is adopted to quantize the output of encoder. In low bit rate image compression, experiments show that our method significantly outperforms JPEG and JPEG2000 and can produce a better visual result with sharp edges, rich textures, and fewer artifacts.

## CCS Concepts

• **Computing methodologies** → **Machine learning** → **Machine learning approaches**

## Keywords

Lossy image compression; U-net like structure; Densely AutoEncoders; convolutional neural networks (CNNs)

## 1. INTRODUCTION

Image compression, which is a type of data compression applied to digital images, has been a fundamental research topic for many decades. Recent image techniques produce very large amounts of data, which may make it prohibitive to storage and communications of image data without the use of compression. Traditional image compression algorithms, such as JPEG [1] and JPEG2000 [2], rely on the hand-crafted encoder/decoder (codec) block diagram. These traditional methods always use the fixed transform matrixes, i.e. Discrete cosine transform (DCT) and wavelet transform, together with quantization and entropy coder to compress the image. As a result, these commonly used method of lossy compression for digital images always introduce the compression artefacts, especially when used at low bit rates. On the other hand, in response to the needs of new media formats, ever-changing hardware technologies, various requirements and content types, compression algorithms need to be more flexible than existing codec. Therefore, the traditional compression methods may not be expected to be optimal and flexible image coding solutions.



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**Figure 1. the JPEG-coded image(right), where we could see blocking artifacts.**

Recently, deep learning and convolutional neural networks (CNNs) have achieved great success in versatile vision tasks and is gradually being used in image compression. Unfortunately, lossy compression is an inherently non-differentiable problem, quantization is a non-differentiable part of the compression pipeline, which makes it difficult to train networks for this task. To solve this problems, several deep learning methods are proposed. Lucas et al. [1] proposed a compressive autoencoder (CAE) structure, together with a differentiable approximation for quantization and entropy rate estimation for an end-to-end training with gradient backpropagation. Toderici et al. [2] proposed a general framework of variable rate image compression and an architecture based on recurrent neural network (RNN). Both above algorithms are representative deep learning methods for lossy image compression.

Based on these existing works, we can find that deep learning methods have great potential in the field of image compression problems, which may be helpful to solve the artefact problems caused by traditional lossy image compression methods. Among these kinds of deep learning methods, autoencoder-based structures have the potential to address the increasing need for flexible lossy compression algorithms. However, the high distortion problem is still remained when used at a low bit rates, which means the existing work are not suitable for image compression and there is still room for improvement.

In this paper, motivated by the excellent performance of CNNs structure, we propose a Densely AuotEncoder (DAE) based lossy image compression architecture, based image compression scheme that optimizes the end-to-end distortion performance of image compression. Overall structure is consisted of a forward encoder, a binarizer, and a backward decoder. My main contributions are as follows:

1) To replace the transform and inverse transform in traditional codecs, we design a symmetric densely autoencoder (DAE)-based architecture with several densely downsampling blocks and densely upsampling blocks to make full use of the feature information obtained from image. Inspired by DenseNet[3], a well-known convolutional neural network structure, we design a new kind of densely connection to get richer feature information from image.
2) To reduce the distortion rate caused by the quantization during the compression processing, we design a U-net like structure with reference to U-Net, which is a famous encoder-decoder structure in the field of image segmentation, so that we can make sure the decoder part can retain as much information as possible from encoder part.
3) A binarizer from Toderici et al. [4] is adopted to quantize the output of encoder. Furthermore, we improve the binarizer part based on the existing work, we add a loss function for binarier part to provide regularization so that the network can have a better robustness.
4) To show our method is more applicable to other existing codecs (e.g. JPEG, JPEG2000), we designed a comparison experiment to compare with the existing codec. We use a subset of CLIC dataset to train and test my network. The experiment results show that my method are more applicable to other tasks.

To sum up, the main contribution of this paper is to introduce the densely autoencoder structure for image compression. This DAE can not only decrease the distortion rate caused by the quantization part of the traditional method, but also enhance the reconstruction image during the decoder part.

The rest of this paper is organized as follows. Section 2 presents a brief review of related work. Section 3 elaborates the proposed compression framework, including the detail architectures of the densely autoencoders, the U-net like structure, and the binarizer used for quantization. Section 4 illustrates the training parameters setting and the solutions to train the whole network. Experimental results are also reported in Section 4. In Section 5, I conclude this paper.

## 2. RELATED WORK

Traditional image compression algorithms, e.g., JPEG and JPEG 2000, rely on the hand-crafted encoder/decoder (codec) block diagram. In the encoding stage, they first perform a linear transform to an image. After that, quantization and entropy coding are used for compressing the image. Using JPEG as an example, JPEG applies discrete cosine transform (DCT) on $8\times8$ image patches as a linear transform, and then quantizes the frequency components by a rounding function and compresses the quantized codes with a variant of Huffman encoding. As for JPEG 2000, it uses a multi-scale orthogonal wavelet decomposition to transform an image, and together with the Embedded Block Coding with Optimal Truncation to encode the image. In the decoding stage, decoding algorithm and inverse transform are designed to minimize distortion. Right now, JPEG and JPEG2000 are commonly used image compression method for storage and transmission of images. However, when used at a low bit rates, traditional methods are not satisfactory, the high distortion rates makes it difficult for them to obtain high quality compression results.

Recently, several deep learning-based image compression models have been proposed. For lossy image compression, several methods have achieved competitive performance. Toderici et al. [2] proposed a general framework of variable rate image compression and an architecture based on recurrent neural network (RNN). They also used a stochastic binarization function to do quantization. Furthermore, they introduce a set of full-resolution compression methods for progressive encoding and decoding of images [4]. These methods learn the compression models by minimizing the distortion for a given compression rate.

Theis et al. [1] propose the famous compressive autoencoders (CAE) structure, they used several residual blocks to build the autoencoder. Further5ore, they compare several approximation functions of the quantization part which are commonly used in CNN compression structures and design a new smooth approximation of the discrete of the rounding function. They use Gaussian scale mixtures as a model for the distributions of quantized coefficients. They also upper-bound the discrete entropy rate loss for continuous relaxation.

Li et al. [5] introduce the importance map into lossy image compression at the first time. Since the local information content is spatially variable in the image, they think that the bit rates of different parts of the image should be adapted to the local content. They use content-weighted importance maps to guide the allocation of content-aware bit rates. Therefore, the sum of importance maps can be used as a continuous alternative to discrete entropy estimation to control the compression rate. And binarizer is adopted to quantize the output of encoder due to the binarization scheme is also directly defined by the importance map.

Jiang et al. [6] design a new kind of end-to-end compression framework based on convolutional neural network. A CNN structure, named compact convolutional neural network (ComCNN), learns an optimal compact representation from an input image, another CNN structure, named reconstruction convolutional neural network (RecCNN), is used to reconstruct the decoded image with high-quality in the decoding end, together with an image codec (e.g., JPEG, JPEG2000 or BPG), a complete end-to-end structure is formed. Furthermore, they develop a unified end-to-end learning algorithm to simultaneously learn ComCNN and RecCNN, which also let the whole structure can be trained in an end-to-end way.

In addition to the above compression algorithms, there are some image super-resolution (SR) method which also work well in image compression. Dong et al. propose a CNN based SR method [15] named SRCNN which consists of three layers: patch extraction, non-linear mapping and reconstruction. There are other SR methods with deeper layers which also work well.

Overall, although the image compression methods based on deep learning achieve good performance, there are still some shortcomings, especially when used at low bit rates, which may limit their application. In contrast, the proposed compression framework tries to solve this problem, by using the densely autoencoder-based structure, the U-Net like structure and the binarizer, the proposed method can get a better compression performance.

## 3. THE PORPOSED METHOD

In this section, we first introduce the architecture of the proposed compression framework and then present the detailed learning algorithm.

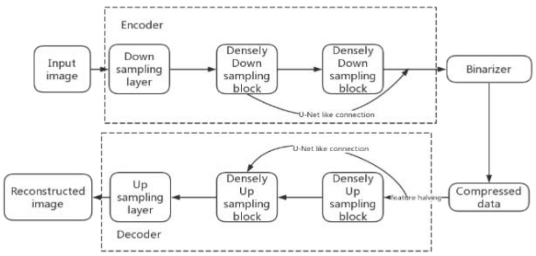


**Figure 2. The proposed DAE compression framework**

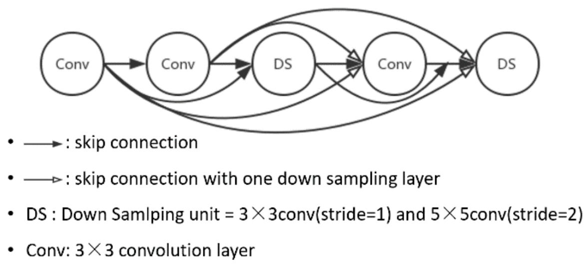


**Figure 3. The structure of the densely downsampling block.**

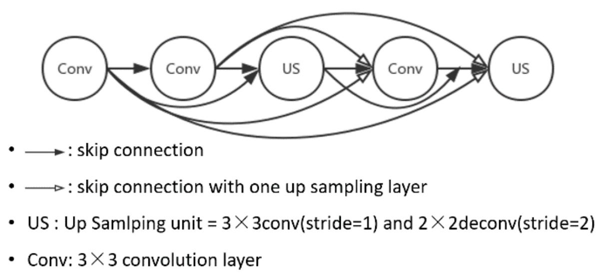


**Figure 4. The structure of the densely upsampling block.**

### 3.1 Architecture of Densely AutoEncoder

As shown in Figure 2, the proposed compression framework consists of three parts: the encoder part, the decoder part and the binarizer part. The encoder part is used for generating a compact representation of the input image for the encoding, binarizer is used for quantifying the output of encoder, the decoder part, which is the mirror symmetry of encoder part, is used for reconstructing the image based on the binary output from the binarizer.

For encoder part, a downsampling layer is first used for reducing the size of the feature map so that we can speed up computation, the downsampling layer we used here is just a $5 \times 5$ convolutional layer with stride of 2. After that, followed by two densely downsampling blocks, the detail information of the densely downsampling blocks is shown in Figure 3. Conv means $3 \times 3$ convolution layer, DS means one down sampling unit which is composed of one $3 \times 3$ convolution layer and one $5 \times 5$ convolution layer with stride of 2. Three convolutional layers and two downsampling units are connected together by the proposal densely connection.

For decoder part, the decoder is just the mirror symmetry of encoder part, two densely upsampling blocks is used, which are just the mirror symmetry of the two densely downsampling blocks used in encoder, and finally one upsampling layer is used for restoring the image to its original size. The structure of the densely upsampling block is shown in Figure 4, Conv also means $3 \times 3$ convolution layer and US means up sampling unit which is composed of one $3 \times 3$ convolution layer and one $2 \times 2$ convolution layer with stride of 2. In decoder, we use the sub-pixle layer[14] as the upsampling layer, which is inspired by the work of Shi et al[7]. Similar to encoder, the proposal densely connection is used to connect all the layers together.

As for binarizer part, it is used for quantifying the output of encoder. To provide regularization, this paper adds a loss function for the binarizer. The detail about the binarizer will be introduced later.

### 3.2 Densely Connection

Generally, deep learning compression methods use neural network to replace transform and inverse transform in traditional codecs because they think neural network can get a better representation of the input image and it's good for compression. However, for existing works, the network may be too simple to make full use of feature information. To solve this problem, I design the densely downsampling block and the densely upsampling block which are shown in Figure 3 and Figure 4. Inspired by the famous DenseNet[3], I use these kinds of densely connection to connect all layers directly with each other so that we can ensure maximum information flow between layers in the network. To preserve the feed-forward nature, each layer obtains additional inputs from all preceding layers and passes on its own feature-maps to all subsequent layers. By preserving the feature information extracted from earlier layers we can retain richer feature information, we can get a better representation of the image, which is obviously helpful to the compression of the image. Densely connection also requires fewer parameters than traditional convolutional autoencoder structures, as there is no need to relearn redundant feature-maps.

In addition, we can also introduce $1\times1$ convolution as bottleneck layer before each $3\times3$ convolution in the densely downsampling block and the up sampling block to reduce the number of input feature-maps, and thus to improve computational efficiency.

Furthermore, compared to existing algorithms, the proposal method combines the convolutional layer and the downsampling layer together as a whole sub-structure for the first time. The convolution layers are used for extracting a better representation of the image, and the downsampling layers are used for compression the image based the representation extracted by convolution layers, they should complement each other and coordinate their work.

To use the densely connection, we should make sure all the feature have the same size during the process of the skip connection so that we can do the concatenation. Therefore, two different skip connections are used in the proposal method. One is the original skip connection (solid line shown in Figure 3 and Figure 4), another is the skip connection with one downsampling

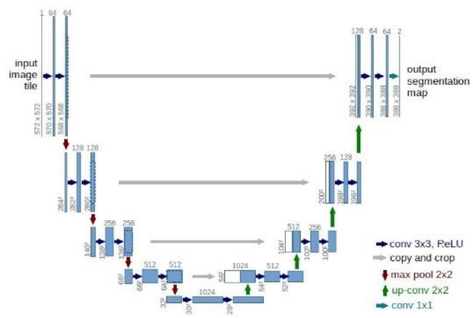


**Figure 5. The U-Net[10] framework**

layer (or one upsampling layer) (hollow line shown in Figure 3 and Figure 4).

Finally, symmetrical encoder and decoder form the main structure of the proposal method, the densely encoder can make sure that even the feature information from earlier layers can be retained, and the symmetric decoder can perfectly reconstruct the image.

## 3.3 U-Net Like Structure

As we know, quantization function is one of the main causes of distortion in image compression algorithm. Because of rounding function, some details in the image are lost when the image is compressed and quantized into binary data., which causes the decoder to be unable to restore the image. If we can retain as much feature information as possible from the encoder during the decoding process, it may help us alleviate the above problems.

U-Net like structure is a good idea to solve the distortion problem caused by the quantization part of the tranditional compression method, it is obvious that the more information retained by encoder, the better compression quality got by decoder. However, for image compression, the encoder part and the decoder part must be separated, so the U-net structure can not be used directly. A simple idea is shown in Figure 3.1, adding the skip connection in encoder part and concatenate the feature together before they enter the binarizer part, and after the binarizer, the feature information from different layers of the encoder can be approximately obtained by simply halving them. And then, by connecting these features to the corresponding layer of the decoder part, the skip connection of U-Net can be imitated approximately.

By using the above way, we simulate U-Net approximately to realize the skip connection from the encoder to the corresponding layer of the decoder. It ensures that feature information from even the earlier layers of the encoder can be retained as much as possible. Experiment results show that even if binarizer's quantification process causes some loss of these feature information, they still contribute to the improvement of compression quality.

Furthermore, by simulating U-Net approximately to realize the skip connection from the encoder to the corresponding layer of the decoder, the network is allowed to propagate context information to the decoder part. As is known to all, decoder is used for reconstructing the image based on the compressed data, this kind of context information can help the decoder get higher quality reconstruced image.

Another advantage is similar to the densely connection, the feature from the earlier layers of the encoder can be used to the maximum extent possible, and the corresponding connection in decoder can help to improve the quality of the reconstructed image. Finally, as a U-Net like structure, the proposed structure also has the advantage similar to U-Net: the network structure can be trained end-to-end from very few images dataset while ensuring the compression quality.

## 3.4 Binarizer

Furthermore, this thesis designs a binarizer to quantize the output of the encoder part. Similar to the previous work of Toderici et al. [4], the binarizer is used for quantization. The input of the binarizer, marked as x, is the output feature vector from the encoder part, as the output of the neural network, x obviously consists of continuous floating point numbers. The output of the binarizer, marked as B(x), is the final compressed data after the quantization step provided by binarizer, and B(x) is composed of discrete binary values.

The binarizer consists of two parts. The first part includes generating the required number of outputs (equal to the required number of output bits) in a continuous interval [-1,1], in this case, tanh activations are used. The second part quantizes them and producing a discrete output in the set$\{-1,1\}$. They also use the regularization provided by a loss function to allow they to cleanly backpropagate gradients through this binarization layer.

Therefore, the full binary encoder function is:

$$\mathrm{B(x)} = \mathrm{b(tanh(x))}$$

where x shows the output from encoder and b() means the quantization part which is user for producing the discrete output, and b() is as follow:

$$b = \begin{cases} -1, & \textit{if the input is less than } 0 \\ 1, & \textit{otherwise} \end{cases}$$

Furthermore, to provide the regularization during the training process, this thesis adds a loss function for the binarizer part as follow:

$$Binarizer_{loss} = \mathrm{MSEloss(b(tanh(x)), tanh(x))}$$

The forward pass shows as above. For the backward pass of back-propagation, this thesis takes the derivative of the expectation. Since$E[b(x)] = x$ for all $x \in [-1,1]$, they pass the gradients through b unchanged.

Analysis of the above proposed binarizer. Based on this existing binarizer proposed by Toderici et al. [4], which is widely used in image compression, this thesis makes a further improvement. In Toderici's structure, the randomized quantization which is used for providing regularization makes the output of the encoder far away from the area near the zero point. Because the closer to the zero, the more random the corresponding binarizer output is. This is called quantization noise.

However, the existing binarizer still has some problems. The randomized quantization is used for providing regularization, and it sure can make the network more robust. Meanwhile, it may let the network be more difficult to train, especially when the network structure is complicated or the dataset is too complicated .

In the proposed work, the loss function will make the output of the encoder to be as far as possible from the area near zero, which is similar to the randomized quantization, and it can also provide regularization and it is easier to be trained than the randomized way.

# 4. EXPERIMENTS

To evaluate the performance of the proposed compression method, we conduct experimental comparisons against traditional compression methods JPEG[12] and JPEG2000[13], we also compare our method with a CNN-based method by Ballé[8]. Among the different variants of JPEG, the optimized JPEG with 4:2:0 chroma subsampling is adopted.

## 4.1 Dataset for Training and Testing

Our Deep Densely AutoEncoder compression model is trained on a subset of CLIC2018[11], all the images are high quality lossless PNGs to avoid compression artefacts. We crop these images into 128×128 patches and take use of these patches to train the network. Mean squared error was used as a measure of distortion during training. After training, we test our model on the commonly used Kodak PhotoCD[9] image dataset with the metrics for lossy image compression.

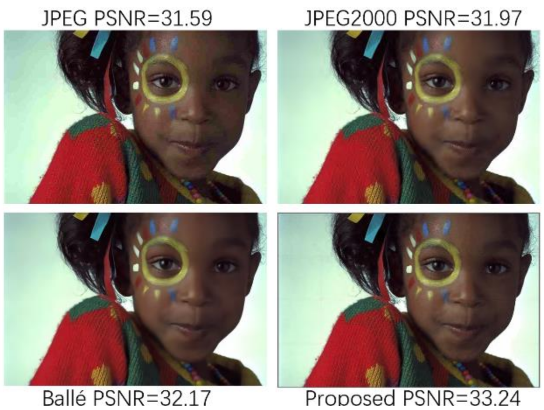


**Figure 6. Compression performance of all competitive algorithms**


**Figure 7. Compression performance of all competitive algorithms**

## 4.2 Experiment Environment and Model Initialization

Our experiments are performed on a PC with 4.20 GHz Intel Core i7-7700K CPU, 16GB RAM and GeForce GTX 1080Ti GPU. We use the Xavier Initialization meothod to initialize the weights of the DAE network and use Adam method with $\alpha = 0.001$, $\beta 1 = 0.9$, $\beta 2 = 0.999$ and $\varepsilon = 10^{-8}$. We train the DAE for 4000 epochs using a batch size of 64. The learning rate is decayed exponentially from 0.001 to 0.00001 for 4000 epochs.

The number of the output channels of the encoder determines the compression ratio. In this paper, we set the number of the output channels of the encoder as 256 to ensure that all the compression methods compress the image with the same bpp.

## 4.3 Experiment Results

The compression rate of our model is evaluated by the metric bits per pixel (bpp), which is calculated as the total amount of bits used to code the image divided by the number of pixels. To ensure fairness, all the compression methods compress the image with the same bpp(0.25)The image distortion is evaluated with Peak Signal-to-Noise Ratio (PSNR), the formula of PSNR is as follows:

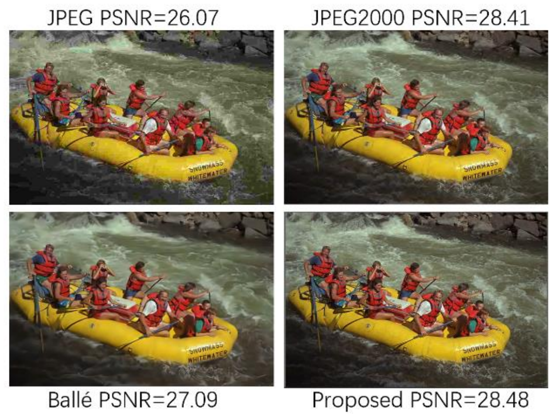


**Figure 8. Compression performance of all competitive algorithms**

**Figure 9. The results of our algorithm have richer image detail, sharper texture and less distortion**

**Figure 10. The results of our algorithm have richer image detail, sharper texture and less distortion**

$$\mathrm{PSNR} = 10 \times log_{10}\left(\frac{(2^n - 1)^2}{MSE}\right)$$

We test all 24 images from Kodak PhotoCD dataset and calculate the average PSNR for JPEG, JPEG2000, Ballé's method and our own structure, the average PSNR result is shown as follows:

**Table 1. Average PSNR result of all competitive algorithms**

| JPEG | JPEG2000 | Ballé[8] | Proposed |
|---|---|---|---|
| 27.03 | 29.59 | 29.76 | 30.14 |

The Figure 6, Figure 7 and Figure 8 show the compression performance of all competitive algorithms.

## 4.4 Experiment Analysis

We use 24 lossless Kodak image to test our method and the experiment results show that at the same compression ratio, our algorithm has higher compression quality, which means the results of our algorithm have richer image detail, sharper texture and less distortion. Figure 6, Figure 7 and Figure 8 further shows that our algorithm has obvious advantages compared to existing algorithms. This is because the densely connection makes the feature information of the image be used to the maximum extent, which is obviously helpful for the encoder to get better representation of the image. Better representation means richer information such as clearer textures, richer details and less blocking artifact.

Furthermore, U-Net like connection provides an approximate skip connection from the encoder to the corresponding layer of the decoder, which ensures the feature information from different layers of the encoder is preserved to the maximum extent. The experimental results are as I analyzed before, U-Net like connection further alleviates the distortion problem.

Finally, an improved binarizer is used for quantification, reasonable regularization method provides high robustness to our algorithm.

Next I will show some details from the results of my compression algorithm. By zooming in on a part of the resulting image, we can clearly see that the results of our algorithm have richer image detail, sharper texture and less distortion.

Just as show in Figure 9 and Figure 10.

# 5. CONCLUSIONS

In this paper, we propose a densely autoencoder based lossy image compression framework. Inspired by DenseNet, which is the famous deep neural network structure, we design a densely downsampling block for encoder and a densely upsampling block for decoder so that we can make full use of the feature information extracted from the input image. Next, inspired by U-Net structure, which is a famous encoder-decoder structure for image segmentation, we design an U-Net like connection so that the decoder part can retain as much information as possible from encoder part.. Furthermore, An improved binarizer is designed to quantize the output of encoder. Experimental results demonstrate that the proposed compression framework is much better than most post-processing algorithms.

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